Q1-A: **Define Symmetric Group S_4 and Show That It Is a Group Under Multiplication**

Radhika Sharma

Preliminaries

Definition 1 (Permutation)**.** *Let $X = \{1, 2, 3, \dots, n\}$ be a finite set. A **permutation** of X is a bijective function $\sigma : X \to X$.*

Definition 2 (Symmetric Group S_n)**.** *The **symmetric group** S_n is the set of all permutations of the set $\{1, 2, \dots, n\}$, equipped with the binary operation of **composition of functions** (also called* multiplication of permutations*).*

$$S_n = \{\, \sigma \mid \sigma : \{1, 2, \dots, n\} \to \{1, 2, \dots, n\} \text{ is a bijection} \,\}$$

The group operation $\circ$ is defined by $(\sigma \circ \tau)(x) = \sigma(\tau(x))$ for all $x \in \{1, \dots, n\}$.

The Symmetric Group S_4

Definition 3 (Symmetric Group S_4)**.** *The **symmetric group** S_4 is the group of all bijections (permutations) from the set $\{1, 2, 3, 4\}$ to itself, under the operation of composition. Formally,*

$$S_4 = \{\, \sigma : \{1, 2, 3, 4\} \xrightarrow{\sim} \{1, 2, 3, 4\} \,\},$$

with $|\, S_4 \,| = 4! = 24$.

Remark 1. The 24 elements of S_4 consist of:

- **1** identity permutation,
- **6** transpositions (2-cycles),
- **3** products of two disjoint transpositions,
- **8** 3-cycles, and
- **6** 4-cycles.

A typical element is written in *two-line notation* or in *cycle notation.* For example,

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} = (1\ 2)(3\ 4).$$

S_4 Is a Group Under Composition

To prove that $(S_4, \circ)$ is a group, we verify the four group axioms: **Closure**, **Associativity**, **Identity**, and **Invertibility**.

Theorem 1. *$(S_4, \circ)$ is a group.*

Proof. **(G1) Closure.**
Let $\sigma, \tau \in S_4$. Then σ and τ are bijections from $\{1, 2, 3, 4\}$ to itself. The composition $\sigma \circ \tau$ is the composition of two bijections, which is again a bijection from $\{1, 2, 3, 4\}$ to $\{1, 2, 3, 4\}$. Hence $\sigma \circ \tau \in S_4$.

Example: Let $\sigma = (1\ 2\ 3)$ and $\tau = (1\ 2\ 4)$ in S_4.

$$(\sigma \circ \tau)(1) = \sigma(\tau(1)) = \sigma(2) = 3, \quad (\sigma \circ \tau)(2) = \sigma(\tau(2)) = \sigma(4) = 4,$$
$$(\sigma \circ \tau)(3) = \sigma(\tau(3)) = \sigma(3) = 1, \quad (\sigma \circ \tau)(4) = \sigma(\tau(4)) = \sigma(1) = 2.$$

So $\sigma \circ \tau = (1\ 3)(2\ 4) \in S_4$. ✓

(G2) Associativity.
For all $\sigma, \tau, \mu \in S_4$ and any $x \in \{1, 2, 3, 4\}$,

$$\big((\sigma \circ \tau) \circ \mu\big)(x) = (\sigma \circ \tau)(\mu(x)) = \sigma(\tau(\mu(x))),$$
$$\big(\sigma \circ (\tau \circ \mu)\big)(x) = \sigma\big((\tau \circ \mu)(x)\big) = \sigma(\tau(\mu(x))).$$

Since function composition is always associative, we have $(\sigma \circ \tau) \circ \mu = \sigma \circ (\tau \circ \mu)$ for all $\sigma, \tau, \mu \in S_4$. ✓

(G3) Existence of Identity.
Define the identity permutation $e : \{1, 2, 3, 4\} \to \{1, 2, 3, 4\}$ by $e(x) = x$ for all x. Clearly e is a bijection, so $e \in S_4$. For any $\sigma \in S_4$ and all $x \in \{1, 2, 3, 4\}$,

$$(\sigma \circ e)(x) = \sigma(e(x)) = \sigma(x), \qquad (e \circ \sigma)(x) = e(\sigma(x)) = \sigma(x).$$

Hence $\sigma \circ e = e \circ \sigma = \sigma$, and e is the identity element in S_4. ✓

(G4) Existence of Inverses.
Let $\sigma \in S_4$. Since σ is a bijection, it has an inverse function $\sigma^{-1} : \{1, 2, 3, 4\} \to \{1, 2, 3, 4\}$ defined by

$$\sigma^{-1}(y) = x \iff \sigma(x) = y.$$

This σ^{-1} is also a bijection, so $\sigma^{-1} \in S_4$. By definition of function inverse,

$$(\sigma \circ \sigma^{-1})(x) = \sigma(\sigma^{-1}(x)) = x = e(x),$$
$$(\sigma^{-1} \circ \sigma)(x) = \sigma^{-1}(\sigma(x)) = x = e(x),$$

so $\sigma \circ \sigma^{-1} = \sigma^{-1} \circ \sigma = e$. ✓

Example: If $\sigma = (1\ 2\ 3\ 4)$, then $\sigma^{-1} = (4\ 3\ 2\ 1) = (1\ 4\ 3\ 2) \in S_4$.

Since all four group axioms are satisfied, $(S_4, \circ)$ is a group. □

Order and Structure

The order of S_4 is $|S_4| = 4! = 24$. It is a **non-abelian** group, since composition of permutations is in general not commutative. For instance, with $\sigma = (1\ 2)$ and $\tau = (1\ 3)$:

$$\sigma \circ \tau = (1\ 3\ 2), \qquad \tau \circ \sigma = (1\ 2\ 3),$$

and $(1\ 3\ 2) \neq (1\ 2\ 3)$, so $\sigma \circ \tau \neq \tau \circ \sigma$.

Hence, S_4 is a **non-abelian group of order 24** under the operation of composition of permutations. $\blacksquare$